

Can Geography Stand In for Labeled Data in Arabic Dialect Identification? A Cross-Linguistic Stress Test of Geographic N -Gram Interpolation

Anonymous ACL submission

Abstract

Arabic dialect identification covers only a fraction of the Arabic speaking world, and most regions have no labeled data at all. A natural starting point is Tobler’s law, which states “*everything is related to everything else, but near things are more related than distant things.*” Applying it to dialects means that nearby dialects tend to be more alike. We test this idea directly for Arabic, then check whether it holds up by repeating the same test on six more language families. For Arabic, we build a model for an unseen dialect from its nearest known dialects, weighted by distance. On the NADI 2020 benchmark across 17 dialects, this recovers **63% of oracle macro F1 with zero target labels**, the best score among all seven families we test. For Mauritania and Somalia, two dialects missing from training entirely, our method still gives real predictions. Repeating the protocol on German, Russian, English, Portuguese, Spanish, and French, we find that recovery is useful everywhere, from 26% to 63%. But the correlation between geography and similarity does not predict which families benefit: neither its size (Spearman $\rho = -0.21$) nor its sign ($\rho = -0.71$, $p = 0.07$) gives a rule a practitioner could trust. The lesson for Arabic dialect identification is good news with a catch. The code available at: <https://anonymous.4open.science/r/counterplay-A37E>

1 Introduction

Arabic dialect identification (DID) assigns informal text to one of the many regional varieties spoken across the Arab world. The task matters for machine translation, sentiment analysis, and information retrieval, where treating Arabic as a single language hides large differences between, say, Peninsular and Maghrebi usage (Zaidan and Callison-Burch, 2014). Resources such as the MADAR corpus (Bouamor et al., 2018) and dialect identification datasets like DSL-TL (Haapanen et al.,

2023) have pushed coverage forward, yet systems exist for only a fraction of Arabic’s dialect continuum: the largest shared benchmark, NADI, covers under 20 of more than 20 dialect-bearing countries, and most regions have no labeled data at all.

The core problem is annotation cost. Building a DID system for a new dialect requires collecting and labeling text from that region, which may be slow, expensive, or impossible for an under-resourced variety, a challenge that mirrors the broader difficulty of language identification for low-resource varieties on noisy web text (Caswell et al., 2020). **Geographic proximity** is a natural guide: if nearby dialects are more similar, a model trained on neighboring regions can stand in for an unseen one. This idea, grounded in Tobler’s first law of geography (Nerbonne and Heeringa, 1997), has direct support for Arabic: Alsudais et al. (2022) found that geographically closer Arabic dialects have more similar n -gram profiles. What is missing is a working system built on that finding, evaluated against a real benchmark, and a sense of how much to trust it.

We supply both. First, we build and evaluate a geographic interpolation system for Arabic dialect identification on the NADI 2020 benchmark, including a true zero shot test on dialects entirely absent from training and an annotation cost analysis that tells practitioners how many labels it takes to beat the zero shot baseline. Second, because a result from one language family could be an accident of Arabic’s particular history rather than a general property of geographic interpolation, we stress-test the method on six more families that spread across the globe in different ways: through **in situ evolution** (communities stay in place and dialects drift through contact with neighbors, as in Arabic itself), **colonial spread** (Spanish, Portuguese, and English carried to the Americas from a recent common European source), **imperial spread** (Russian carried across Central Asia by Soviet rule, native in Rus-

sia but imposed elsewhere), and **mixed** settlement (French, spanning in situ European varieties and colonial African ones). If the geographic–linguistic correlation r that [Alsudais et al. \(2022\)](#) measured for Arabic could be computed on unlabeled data and used to decide, ahead of time, whether geographic interpolation is worth deploying for a new dialect, that would turn their finding into a general recipe. We test whether it does. We find that it does not: r is not a reliable advance signal even though the method itself works well for Arabic.

Our mixture model builds a character n -gram model for an unseen country from its k nearest known countries, weighted by inverse geographic distance between country centroids (Section 4).

Our contributions are:

1. The first system-level evaluation of geographic interpolation for Arabic DID against a real external benchmark (NADI 2020, 17 dialects): 63% of oracle macro F1 recovered with zero target labels, the strongest result of any family we test.
2. A true zero shot evaluation on Mauritanian and Somali Arabic, dialects absent from all training data, where our method is the only one to give non-degenerate predictions, plus an annotation cost analysis showing that around 10 labeled tweets per dialect beat the zero shot baseline.
3. A six-family stress test (German, Russian, English, Portuguese, Spanish, French; 58 further varieties) of whether the Arabic result generalizes as a recipe. Recovery is a usable proxy everywhere (26–63%), but the geographic–linguistic correlation r does *not* predict which families benefit: neither $|r|$ (Spearman $\rho = -0.21$) nor signed r ($\rho = -0.71$, $p = 0.07$) gives a reliable advance rule. Arabic’s success cannot be safely assumed to transfer to the next untested language.

2 Related Work

Dialect and language identification. Character n -gram models have been strong baselines for variety discrimination since [Cavnar and Trenkle \(1994\)](#) and remain competitive ([Tiedemann and Ljubešić, 2012](#); [Zampieri et al., 2014](#)). For Arabic, [Salameh et al. \(2018\)](#) use smoothed n -gram language models for fine grained DID. The NADI series ([Abdul-Mageed et al., 2020, 2024](#)) tests country level Arabic DID including an unseen dialect track. See [Jauhiainen et al. \(2019\)](#) for a full survey.

Geography and dialect variation. Dialectometry ([Nerbonne and Heeringa, 1997](#); [Nerbonne, 2010](#)) measures how dialects vary across space. [Eisenstein et al. \(2010\)](#) showed that Twitter location correlates with word choice, and [Lui and Cook \(2013\)](#) used geotagged n -gram models for English national varieties. For Arabic, [Alsudais et al. \(2022\)](#) found a negative correlation between Haversine distance and dialect similarity and called for using this to transfer models. We test that idea and probe its limits across six more families.

Zero shot cross-dialect transfer. Zero shot cross lingual transfer is well studied for transformers ([Inoue et al., 2021](#)), but geographic structure has not been used as a guide. NADI 2024 ([Abdul-Mageed et al., 2024](#)) added an unseen dialect track, where participants used transformer fine tuning and prompting. This work instead maps when geographic proximity is, and is not, a reliable guide for building dialect models.

Weak geographic supervision. CC-TLD filtering has been used as a noisy geographic label for web text ([NLLB Team et al., 2022](#)). Geotagged social media has been used to study lexical diffusion ([Eisenstein et al., 2010](#)) and national variety differences ([Lui and Cook, 2013](#)). We bring both sources together under a unified evaluation framework, using CC-TLD web text as a similar, if noisier, geographic label for the six families beyond Arabic.

3 Data

3.1 Arabic: Geotagged Twitter

We use geotagged tweets from [Rezk \(2021\)](#): 18 countries, 440,000 tweets, users assigned by their declared Twitter location. Twitter captures *colloquial* Arabic because of diglossia. Arabic has two coexisting registers: Modern Standard Arabic (MSA), used in formal writing and broadcast media, and a set of regional spoken dialects that differ enough to impede mutual comprehension across regions. Because social media is informal, speakers write in their own dialect. Peninsular Arabic, Levantine Arabic, and Maghrebi Arabic are phonologically and lexically distinct in these colloquial texts, making geotagged Twitter a reliable source of dialectal signal.

Language	Source	#var	Docs	Register
Arabic	Twitter	17	440K	colloquial
English	mc4 CC-TLD	11	95K	formal
French	mc4 CC-TLD	13	85K	formal
Spanish	mc4 CC-TLD	17	51K	formal
Portuguese	mc4 CC-TLD	3	65K	formal
Russian	mc4 CC-TLD	9	75K	formal
German	mc4 CC-TLD	5	38K	formal

Table 1: Dataset overview. #var = varieties scored in the leave one out macro (oracle F1 >0.01). Arabic trains 18 and Spanish trains 19, but Bahrain, Panama, and Paraguay have near-zero oracle and are dropped. Docs = total documents used (capped at 10,000 per variety for CC-TLD languages). Arabic uses geotagged Twitter. All other languages use CC-TLD filtered web text from mc4.

3.2 CC-TLD Web Text

For all other six languages, we use country-code top-level domain (CC-TLD) filtered documents from the mc4 multilingual web corpus (NLLB Team et al., 2022). Each document is assigned to a country based on its URL domain: a document from a .de domain is assigned to Germany, one from .mx to Mexico, and so on. We collect up to 10,000 documents per country, cut each to 500 characters for efficiency, and remove any country with fewer than 80 documents.

CC-TLD is a noisy signal: a .de domain does not guarantee native German content, and diaspora communities produce off-country text. Despite this noise, the aggregate signal across thousands of documents is strong enough to produce distinct character n -gram profiles per country. Unlike geotagged Twitter, mc4 web text leans toward *formal written* language, which turns out to matter for whether geographic structure is present. We return to this in Section 6.2.

3.3 Dataset Overview

Table 1 summarizes the data for all seven language families. Arabic varieties are trained on the full available data. CC-TLD varieties are capped at 10,000 documents each. Country centroids are computed as population-weighted geographic mid-points, taken from public geographic databases.

4 Method

4.1 Character n -Gram Language Models

We model each geographic variety as a character n -gram language model with interpolated Kneser-Ney smoothing (Chen and Goodman, 1999), order

$n=4$. Character n -grams capture subword patterns that differ across dialects: spelling conventions, affixes, clitics, and script-specific ligatures. They are script agnostic and require no tokenizer, making them easy to deploy across the seven languages in this study. One model is trained per variety from that variety’s documents.

At test time, text t is assigned to the variety with the highest log-linear score:

$$\hat{c} = \arg \max_c [\log P(t | c) + \log P(c)], \quad (1)$$

where $P(c)$ is a prior over varieties estimated from training document counts.

4.2 Geographic Interpolation and Correlation

For an unseen country c^* , we build a synthesized model as a weighted mixture over the $k=5$ nearest known countries:

$$\hat{P}(w | c^*) = \sum_{i \in \mathcal{N}_k(c^*)} \alpha_i P(w | c_i), \quad (2)$$

with $\alpha_i \propto 1/d(c^*, c_i)$, where $d(\cdot, \cdot)$ is the Haversine distance between country centroids. The interpolated cell takes the average log prior of its source countries. This requires no labeled data from c^* , only its location.

Whether this works depends on whether geographic distance correlates with linguistic similarity. We measure this via the Pearson correlation r over all $\binom{V}{2}$ country pairs in a family. For each pair (c_i, c_j) we compute the Haversine distance d_{ij} and the cosine similarity s_{ij} between n -gram probability vectors:

$$s_{ij} = \frac{\mathbf{p}_i \cdot \mathbf{p}_j}{\|\mathbf{p}_i\| \|\mathbf{p}_j\|}, \quad r = \text{Pearson}(\{d_{ij}\}, \{s_{ij}\}), \quad (3)$$

where $\mathbf{p}_c$ is the vector of n -gram log probabilities under model c . Negative r means closer varieties are more similar (Tobler’s dialectal law holds). Positive r means more distant varieties are more alike. Near-zero r means distance carries no information about similarity.

4.3 Evaluation Protocol

Leave one out (LOO). For Arabic, we hold out each of 18 countries in turn and test the full 18-class classifier on the NADI 2020 development set (4,957 tweets, external benchmark). For all other languages, we use intrinsic LOO: each country’s documents are split 80% train / 20% test, and we

Lang.	Type	r	#v	Oracle	Rec.
Arabic	in situ colloquial	-0.43	17	0.193	63%
German	in situ written	+0.30	5	0.704	57%
English	colonial spread	+0.03	11	0.532	48%
Russian	imperial spread	+0.33	9	0.755	45%
French	mixed EU+Africa	-0.14	13	0.758	45%
Portug.	colonial (few)	+0.50	3	0.699	33%
Spanish	colonial (many)	+0.37	17	0.368	26%

Table 2: Cross-language LOO summary. r = Pearson correlation of pairwise cosine similarity vs. Haversine distance (Eq. 3). Negative r means closer varieties are more similar. Rec. = geo macro F1 / oracle macro F1. Sorted by recovery.

test the held out country against a classifier trained on the rest.

We report macro F1 in two conditions. The *oracle* classifier is trained with the held out country’s own data (the upper bound). The *geo* classifier replaces the held out country with the geographic mixture from Eq. 2 (the zero shot system). **Recovery** = geo macro F1 / oracle macro F1 measures how much of the oracle performance the zero shot system achieves.

True zero shot. For Mauritania and Somalia, present in NADI 2020 but absent from our Arabic training data, we add a built cell to the 18-cell system and test directly.

Annotation cost. We train supervised n -gram models on N labeled Arabic tweets per variety (varying N) and compare against the zero shot baseline to find the crossover point.

5 Results

5.1 Cross-Language Summary

Table 2 shows the LOO results for all seven language families alongside the geographic-linguistic correlation r .

Two things stand out. *First*, the method is a usable proxy everywhere. Recovery ranges from 26% to 63% of oracle with no target labels, and no family fails outright. *Second*, r does not order the families. Arabic ($r=-0.43$) and Portuguese ($r=+0.50$) have the two largest $|r|$ values, yet sit at opposite ends of the recovery range (63% and 33%). English ($r=+0.03$, almost no correlation) recovers 48%, more than Russian and Spanish, which have much larger $|r|$. The one weak trend is by sign, not magnitude: the two families with $r<0$ (Arabic, French) land above the positive ones on average.

Section 6.1 shows that neither reading holds up statistically.

5.2 Correlation Plots

Figure 1 plots pairwise cosine similarity of n -gram profiles against Haversine distance for all seven language families.

Arabic’s negative correlation ($r=-0.43$, $p<0.001$) confirms Alsudais et al. (2022): nearby Arabic varieties are genuinely more alike. The geotagged Twitter data captures colloquial speech, where dialectal signal is strong.

German’s correlation is positive but modest ($r=+0.30$). Germany, Austria, and Switzerland are all in situ Germanic varieties that have changed together over centuries. The data source matters here: mc4 web text tends toward *Standard German* regardless of country, which flattens the regional features that distinguish Bavarian, Alemannic, and Swiss German in spoken language.

French ($r=-0.14$, $p>0.05$) and English ($r=+0.03$) show almost no relationship. The French family covers European varieties (FR, BE, CH, LU), North African ones (MA, DZ, TN), West African ones (SN, CI, CM, CD), and Canadian. Distance tells us little here, because a European and African pair at 5,000 km can be more alike than two African pairs at 1,000 km. Yet, as the next sections show, both families still recover 45–48% of oracle. Weak correlation does not mean the proxy is useless.

5.3 Arabic: In Situ with Colloquial Data

Table 3 shows per country Arabic LOO. Geographic interpolation reaches macro F1 of **0.121** against an oracle of **0.193** (63% recovery).

Recovery varies widely and tracks geographic embeddedness. The Peninsular and Levant cluster recovers best. UAE (129%), Saudi Arabia (123%), and Syria (117%) even beat their own oracle, because their Twitter training data is noisier than their neighbors’ and the mixture borrows cleaner signal. Saudi Arabia and Iraq (98%) are surrounded by near-identical varieties. Lebanon (25%) is a clear exception. It sits only 232 km from Palestine yet recovers poorly. Centuries of Phoenician, Aramaic, and French contact have pulled Lebanese Arabic away from its geographic position. Sudan (6%) is the most isolated case. Its nearest model country (Egypt, 1,258 km) competes rather than helps. This within-Arabic spread, from 6% to over 100%, is wider than the spread *across* whole families, a

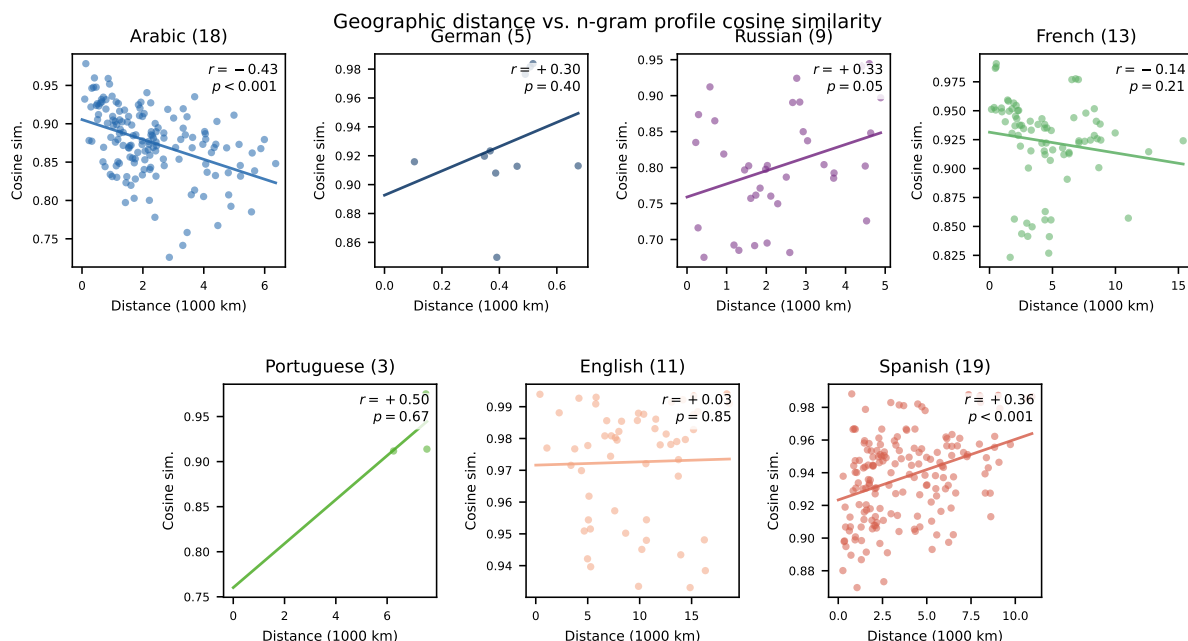


Figure 1: Geographic distance vs. n -gram profile cosine similarity across seven language families. Arabic ($r=-0.43$) and French ($r=-0.14$) show $r<0$ (closer varieties more similar). The rest show $r>0$ or $r\approx 0$. Panel counts include all varieties with enough training data.

first sign that geographic distance alone is a coarse guide.

5.4 Other Language Families

Tables 4–9 show per country LOO for German, Russian, English, French, Portuguese, and Spanish.

German (57%). The best non-Arabic recovery, 45–68% across five compact Central European varieties. Germany itself reaches 68%. The variety set is coherent: mc4 documents from Germany, Austria, and Switzerland all lean toward written Standard German, so any neighbor is a decent stand-in.

Russian (45%). Former Soviet varieties recover at a moderate, even level (33–61%). Russia itself reaches 61% and Georgia 54%, with no obvious advantage for the geographically tight Caucasus cluster (Armenia, Azerbaijan around 48%) over the distant ones. Distance does not order the outcomes.

English (48%). Eleven colonial varieties recover evenly (31–65%). Britain (65%) and Nigeria (61%) recover best, South Africa (31%) least. There is no clean split between the Anglosphere core and the African or Asian varieties, which the near-zero correlation ($r=+0.03$) already foreshadows.

French (45%). Despite the most diverse variety set, French recovers as well as English (39–63%). European French (FR 63%) tops the list and the

African varieties (CI 52%, CM 53%, SN 48%) are not far behind. The earlier intuition that mixing European and African French should fail does not hold once the classifier is scored correctly.

Portuguese (33%). Only three varieties, and the worst recovery of the six, which is the sharpest counterexample to the $|r|$ story: Portuguese has the largest $|r|$ of all (0.50) yet the second lowest recovery. Angola recovers 67% but Brazil and Portugal only 26% each, because Brazil dominates any mixture it joins and is itself distinct from the European standard.

Spanish (26%). Seventeen colonial varieties, the lowest macro recovery. Outcomes range from 0% (three small Central American varieties the mixture never predicts) to 54% (Argentina), with no link to distance.

5.5 True Zero Shot: Varieties Absent from Training

Mauritania and Somalia appear in NADI 2020 but are absent from our Arabic training data. Adding synthesized cells, Mauritania reaches a geo F1 of 0.018 (40 test items, 1,334 km from Morocco) and Somalia 0.025 (51 items, 1,183 km from Yemen). Low F1 is expected at these distances, but both cells give nonzero predictions where fastText (Joulin et al., 2017) and GlotLID (Kargaran et al., 2023)

Country	Oracle	Geo-5	Dist.	Rec.
UAE (AE)	0.097	0.125	323 km	129%
Saudi Arabia (SA)	0.210	0.259	602 km	123%
Syria (SY)	0.095	0.111	293 km	117%
Iraq (IQ)	0.232	0.227	511 km	98%
Morocco (MA)	0.123	0.113	947 km	92%
Algeria (DZ)	0.267	0.225	947 km	84%
Tunisia (TN)	0.140	0.105	991 km	75%
Yemen (YE)	0.153	0.115	989 km	75%
Palestine (PS)	0.098	0.059	126 km	60%
Oman (OM)	0.189	0.108	379 km	57%
Qatar (QA)	0.073	0.040	90 km	55%
Kuwait (KW)	0.074	0.040	465 km	54%
Egypt (EG)	0.567	0.291	689 km	51%
Jordan (JO)	0.111	0.047	126 km	42%
Libya (LY)	0.297	0.107	1123 km	36%
Lebanon (LB)	0.234	0.058	232 km	25%
Sudan (SD)	0.319	0.020	1258 km	6%
Bahrain (BH)	0.000	0.000	90 km	—
Macro[†]	0.193	0.121	—	63%

Table 3: Arabic LOO on NADI 2020 dev (4,957 tweets, 18 countries). Dist. = Haversine to nearest training country. Rec. = Geo-5 / Oracle. [†] Macro excludes Bahrain (oracle ≈ 0). Shading: $\geq 95\%$, 60–94%, 35–59%, $< 35\%$.

CC	Oracle	Geo	Dist.	Rec.
AT	0.711	0.375	387 km	53%
CH	0.659	0.412	104 km	63%
DE	0.610	0.414	462 km	68%
LI	0.737	0.427	104 km	58%
LU	0.805	0.364	368 km	45%
Macro	0.704	0.399	—	57%

Table 4: German per country LOO (CC-TLD intrinsic evaluation). Dist. = Haversine to nearest training country. Shading: $\geq 95\%$, 60–94%, 35–59%, $< 35\%$.

assign all Arabic text to a single undifferentiated label.

5.6 Annotation Cost

Figure 2 shows supervised macro F1 on Arabic NADI 2020 dev as a function of labeled tweets per variety.

Around 10 labeled tweets per variety beat the zero shot baseline on covered varieties. Full oracle performance approaches at $N=1,000$. Geographic interpolation matters most for varieties with no labeled data at all, the exact setting where annotation is most expensive or impossible.

CC	Oracle	Geo	Dist.	Rec.
AM	0.752	0.355	221 km	47%
AZ	0.824	0.408	221 km	49%
BY	0.728	0.242	579 km	33%
GE	0.742	0.398	279 km	54%
KG	0.838	0.396	927 km	47%
KZ	0.759	0.282	927 km	37%
MD	0.863	0.354	291 km	41%
RU	0.561	0.345	3041 km	61%
UA	0.725	0.303	291 km	42%
Macro	0.755	0.343	—	45%

Table 5: Russian per country LOO (CC-TLD intrinsic evaluation). Dist. = Haversine to nearest training country. Shading: $\geq 95\%$, 60–94%, 35–59%, $< 35\%$.

CC	Oracle	Geo	Dist.	Rec.
AU	0.416	0.180	4155 km	43%
CA	0.480	0.236	2257 km	49%
GB	0.404	0.261	403 km	65%
GH	0.633	0.339	1075 km	54%
IE	0.522	0.270	403 km	52%
IN	0.593	0.213	5010 km	36%
KE	0.619	0.324	3388 km	52%
NG	0.650	0.394	1075 km	61%
NZ	0.498	0.227	4155 km	46%
US	0.568	0.245	2257 km	43%
ZA	0.465	0.145	3753 km	31%
Macro	0.532	0.258	—	48%

Table 6: English per country LOO (CC-TLD intrinsic evaluation). Dist. = Haversine to nearest training country. Shading: $\geq 95\%$, 60–94%, 35–59%, $< 35\%$.

6 Analysis

6.1 Does r Predict Recovery? No.

The natural hope is that r , which a practitioner can compute from unlabeled data alone, tells us in advance whether interpolation is worth deploying. It does not. Table 10 sorts the families by $|r|$. The recovery column does not follow.

Magnitude fails. The two families with the largest $|r|$, Portuguese (0.50) and Arabic (0.43), sit at opposite ends of the recovery range (33% and 63%). English has almost no correlation ($|r|=0.03$) yet recovers 48%, more than Russian, Spanish, and Portuguese, which all have much larger $|r|$. The rank correlation between $|r|$ and recovery is $\rho=-0.21$ ($p=0.65$), indistinguishable from zero and, if anything, pointing the wrong way. There is no $|r|$ threshold a practitioner could use.

Sign gives only a weak hint. The signed correlation does slightly better. The two families with $r<0$, Arabic and French, both land in the upper

CC	Oracle	Geo	Dist.	Rec.
BE	0.574	0.273	159 km	48%
CA	0.764	0.300	8784 km	39%
CD	0.916	0.369	1642 km	40%
CH	0.668	0.303	368 km	45%
CI	0.781	0.410	1253 km	52%
CM	0.776	0.409	1642 km	53%
DZ	0.842	0.328	947 km	39%
FR	0.513	0.325	539 km	63%
LU	0.773	0.307	159 km	40%
MA	0.745	0.360	947 km	48%
MG	0.856	0.331	3088 km	39%
SN	0.847	0.409	1253 km	48%
TN	0.797	0.308	991 km	39%
Macro	0.758	0.341	—	45%

Table 7: French per country LOO (CC-TLD intrinsic evaluation). Dist. = Haversine to nearest training country. Shading: $\geq 95\%$, 60–94%, 35–59%, $< 35\%$.

CC	Oracle	Geo	Dist.	Rec.
AO	0.381	0.256	6251 km	67%
BR	0.874	0.228	7511 km	26%
PT	0.843	0.218	6251 km	26%
Macro	0.699	0.234	—	33%

Table 8: Portuguese per country LOO (CC-TLD intrinsic evaluation). Dist. = Haversine to nearest training country. Shading: $\geq 95\%$, 60–94%, 35–59%, $< 35\%$.

half of the recovery range, which matches the original Tobler intuition that negative r (nearby varieties more similar) is the helpful case. But the trend is $\rho = -0.71$ at $p = 0.07$, not significant at the 0.05 level with seven families, and German breaks it badly: $r = +0.30$ yet 57% recovery, the second best. The honest reading is that r carries a faint directional signal and no usable magnitude signal.

Why the intuition fails. Recovery depends on how the held-out variety sits relative to its neighbors, which is a local property. The correlation r is a single global summary over all $\binom{V}{2}$ pairs. A family can have a flat global r yet still contain neighbor pairs that transfer well, as English and French show. Conversely a high $|r|$ can come from a few distant pairs while the actual nearest neighbors transfer poorly, as in Portuguese, where Brazil dominates every mixture. One number over the whole family cannot capture this.

6.2 Register as a Confound

The Arabic and German contrast isolates the effect of **register**. Both are in situ language fami-

CC	Oracle	Geo	Dist.	Rec.
AR	0.459	0.249	762 km	54%
BO	0.208	0.094	958 km	45%
CL	0.580	0.208	762 km	36%
CO	0.529	0.082	833 km	16%
CR	0.154	0.034	355 km	22%
CU	0.488	0.050	959 km	10%
DO	0.422	0.109	959 km	26%
EC	0.400	0.131	833 km	33%
ES	0.399	0.142	6728 km	36%
GT	0.054	0.000	263 km	0%
HN	0.263	0.000	278 km	0%
MX	0.492	0.050	1560 km	10%
NI	0.245	0.086	278 km	35%
PE	0.548	0.089	896 km	16%
SV	0.182	0.000	263 km	0%
UY	0.392	0.143	1043 km	36%
VE	0.443	0.155	875 km	35%
Macro	0.368	0.095	—	26%

Table 9: Spanish per country LOO (CC-TLD intrinsic evaluation). Dist. = Haversine to nearest training country. Macro excludes countries with oracle ≤ 0.01 (Panama, Paraguay). Shading: $\geq 95\%$, 60–94%, 35–59%, $< 35\%$.

Language	$ r $	r (sign)	Recovery
Portuguese	0.50	+	33%
Arabic	0.43	—	63%
Spanish	0.37	+	26%
Russian	0.33	+	45%
German	0.30	+	57%
French	0.14	—	45%
English	0.03	+	48%

Table 10: Absolute geographic–linguistic correlation $|r|$ and macro LOO recovery, sorted by $|r|$. Spearman($|r|$, recovery) = -0.21 , $p = 0.65$. Spearman(signed r , recovery) = -0.71 , $p = 0.07$. Neither gives a usable rule.

lies with centuries of change among neighboring varieties. Yet Arabic shows negative geographic structure ($r = -0.43$) while German shows positive ($r = +0.30$). The difference is the data source.

Twitter captures *colloquial* Arabic. Because of diglossia, speakers write in their own dialect rather than Modern Standard Arabic. Peninsular Arabic, Levantine Arabic, and Maghrebi Arabic differ greatly in casual speech. mc4 captures *formal written* German. A speaker from Munich and a speaker from Zurich both produce Standard German when writing a blog post or news article. Written standardization removes the regional signal that spoken registers keep.

This sign difference, though, does not translate into a recovery gap. German recovers 57% and Arabic 63%, close despite opposite r . Register shapes

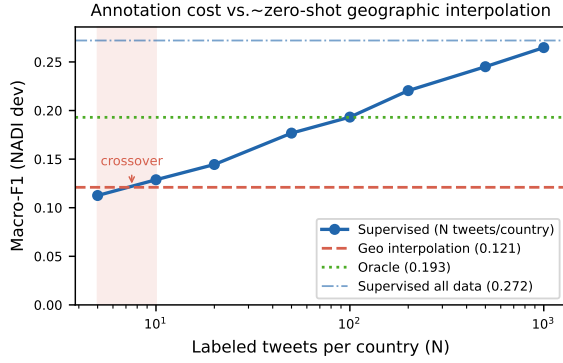


Figure 2: Supervised n -gram macro F1 vs. labeled tweets per Arabic variety (9 varieties with $\geq 1,000$ training examples, means over 3 trials). Dashed: zero shot geo-interpolation baseline (0.121). Dotted: oracle full training (0.193).

where the geographic signal lives, but it does not let r predict how well interpolation will do, which is the broader point of Section 6.1. The practical implication is narrow: colloquial data shows the Tobler pattern more cleanly, but formal data still supports a usable proxy.

7 Discussion

Practical recommendation. For a new dialect variety with no labeled data, geographic interpolation is a reasonable first move. It costs only unlabeled text and a table of centroids, and it recovers 26–63% of oracle macro F1. But do not expect r to tell you in advance how well it will do. We measured r on the same unlabeled data and it did not predict recovery, in magnitude or in sign, strongly enough to act on. Treat interpolation as a cheap baseline to try, not a method you can pre-screen. The cheaper safeguard is a few labels. On Arabic, around 10 labeled tweets per variety already beat the zero shot baseline (Section 5.6), so any variety that can be labeled at all is better served by a small seed set than by interpolation.

Beyond n -gram models. Geographic interpolation is not tied to a particular model type. The same mixture approach (Eq. 2) can be applied to neural language models by interpolating output distributions or by averaging model parameters. Applying it to fine tuned transformer models, such as CAMELBERT (Inoue et al., 2021), is a natural next step for Arabic. Whether the same negative result about r would hold for a neural mixture is an open question this paper does not answer.

Register and contact history. Two descriptive patterns help explain when the proxy works, even though neither rescues r as a screening tool. *Register* shapes where the geographic signal lives: colloquial Twitter keeps the Tobler pattern (Arabic $r < 0$) that Standard German web text flattens ($r > 0$), though both still recover well (Section 6.2). And *contact history can override geography*: Lebanon recovers only 25% despite a neighbor just 232 km away (Section 5.3), a reminder that the mixture in Eq. 2 encodes physical distance only, not the historical contact that actually shapes a dialect.

Future directions. Three extensions seem most direct. The first is testing more language families and more varieties per family, since our central result is a non-correlation that a larger panel could in principle overturn. The second is repeating the same protocol with neural models in place of character n -grams, interpolating output distributions or parameters instead of count-based probabilities, to see whether the same gap between local recovery and global correlation persists once the underlying model is more expressive. The third is moving beyond country centroids toward finer-grained units, such as cities or provinces, which would let the method speak to the within-country variation we set aside in this study. We see this work as a starting point for that agenda, not a closed result.

8 Conclusion

We tested geographic n -gram interpolation for zero shot dialect identification, first on Arabic and then across six more language families, covering 17 Arabic, 17 Spanish, 13 French, 11 English, 9 Russian, 5 German, and 3 Portuguese varieties. The method works as a rough proxy everywhere, recovering 26–63% of oracle macro F1 with no target labels, and does best on Arabic, at 63%.

The main finding, though, is negative: the geographic–linguistic correlation r does *not* predict which families benefit. Magnitude is useless (Spearman $|r|$ vs. recovery $\rho = -0.21$, $p = 0.65$), and sign gives only a weak, non-significant hint that negative r helps ($\rho = -0.71$, $p = 0.07$). A practitioner cannot screen families with r , even though it does help explain Arabic on its own. The broader lesson is cautionary: geography gives a usable but unpredictable proxy for dialect transfer, and a single correlation is not enough to know in advance when it will pay off. A single number, computed once on unlabeled data, is a tempting shortcut, but

it is no substitute for measuring how a method actually performs on the variety at hand.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, Chiyu Zhang, El Moatez Billah Nagoudi, Injy Hamed, Houda Bouamor, and Nizar Habash. 2024. NADI 2024: The fifth nuanced Arabic dialect identification shared task. In *Proceedings of the ArabicNLP Workshop*.

Muhammad Abdul-Mageed, Chiyu Zhang, AbdelRahim Elmadany, Houda Bouamor, and Nizar Habash. 2020. NADI 2020: The first nuanced Arabic dialect identification shared task. In *Proceedings of the Fifth Arabic Natural Language Processing Workshop*, pages 97–110.

Abdulkareem Alsudais, Wafa Alotaibi, and Faye Alomary. 2022. Similarities between Arabic dialects: Investigating geographical proximity. *Information Processing and Management*, 59(1):102770.

Houda Bouamor, Nizar Habash, Mohammad Salameh, Wajdi Zaghouani, Owen Rambow, Dana Abdulmageed, Ossama Obeid, Salam Khalifa, Alexander Erdmann, Nadi Ghazal, and Amr Jardak. 2018. The MADAR Arabic dialect corpus and lexicon. In *Proceedings of LREC*.

Isaac Caswell, Theresa Breiner, Daan van Esch, and Ankur Bapna. 2020. Language ID in the wild: Unexpected challenges on the path to a thousand-language web text corpus. In *Proceedings of COLING*, pages 6588–6608.

William B. Cavnar and John M. Trenkle. 1994. N-Gram-based text categorization. In *Proceedings of SDAIR*, pages 161–175.

Stanley F. Chen and Joshua Goodman. 1999. An empirical study of smoothing techniques for language modeling. *Computer Speech & Language*, 13(4):359–394.

Jacob Eisenstein, Brendan O’Connor, Noah A. Smith, and Eric P. Xing. 2010. A latent variable model for geographic lexical variation. In *Proceedings of EMNLP*, pages 1277–1287.

Laurie Haapanen, Dominik Blanck, Andrea Ceolin, Bjorn Gambäck, and Marcos Zampieri. 2023. DSL-TL: A typologically diverse dataset for dialect and language identification with true labels. In *Proceedings of the Tenth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial)*, pages 12–24.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104.

Tommi Jauhiainen, Marco Lui, Marcos Zampieri, Timothy Baldwin, and Krister Lindén. 2019. Automatic language identification in texts: A survey. In *Journal of Artificial Intelligence Research*, volume 65, pages 675–782.

Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of tricks for efficient text classification. In *Proceedings of EACL*, pages 427–431.

Amir Hossein Kargaran, Ayyoob Imani, François Yvon, and Hinrich Schütze. 2023. GlotLID: Language identification for low-resource languages. In *Findings of EMNLP*, pages 6155–6172.

Marco Lui and Paul Cook. 2013. Classifying English documents by national dialect. In *Proceedings of the Australasian Language Technology Association Workshop*, pages 5–15.

John Nerbonne. 2010. Measuring the diffusion of linguistic change. *Philosophical Transactions of the Royal Society B*, 365:3821–3828.

John Nerbonne and Wilbert Heeringa. 1997. Measuring dialect distance phonetically. In *Proceedings of the Workshop on Computational Phonology*, pages 11–18.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, and 1 others. 2022. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*.

Abdelrahman Rezk. 2021. Arabic dialect identification dataset. https://huggingface.co/datasets/Abdelrahman-Rezk/Arabic_Dialect_Identification. Tweets collected via user self-declaration of country in Twitter profile.

Mohammad Salameh, Houda Bouamor, and Nizar Habash. 2018. Fine-grained Arabic dialect identification. In *Proceedings of COLING*, pages 1332–1344.

Jörg Tiedemann and Nikola Ljubešić. 2012. Efficient discrimination between closely related languages. In *Proceedings of COLING*, pages 2619–2634.

Omar F. Zaidan and Chris Callison-Burch. 2014. Arabic dialect identification. In *Computational Linguistics*, volume 40, pages 171–202.

Marcos Zampieri, Liling Tan, Nikola Ljubešić, and Jörg Tiedemann. 2014. A report on the DSL shared task 2014. In *Proceedings of the First Workshop on Applying NLP Tools to Similar Languages, Varieties and Dialects (VarDial)*, pages 58–67.

Limitations

Arabic data quality. Arabic results rely on geo-tagged Twitter data, which has known biases. Users declare their location manually, and many provide a city or country that does not reflect where they

currently live. Diaspora communities tweet in their heritage dialect from abroad, mixing a Moroccan dialect signal into a Belgian or French IP address. Urban areas are overrepresented because Twitter usage is concentrated in cities, so rural and peripheral dialectal features are undersampled. Code switching between Arabic and French, English, or local languages is common in Maghrebi and Peninsular tweets and adds noise to the n -gram profiles.

CC-TLD as a geographic proxy. The CC-TLD assumption maps a URL domain to a country of origin. This is violated in several ways. Multinational companies host content under a single national domain regardless of the intended audience. Diaspora communities in Germany write in Arabic or Turkish under .de domains. Bilingual countries like Belgium, Canada, and Switzerland produce mixed language content under a single CC-TLD. International organizations and government sites often use national domains while writing in multiple languages. Despite these violations, the aggregate signal across thousands of documents is consistent enough to produce useful n -gram profiles, but individual country profiles may be noisier than the numbers suggest.

Country centroids. We represent each country as a single geographic point. This ignores within country dialect variation: Germany has northern and southern varieties, the United States has regional accents across thousands of kilometers, and Morocco spans Saharan and Atlantic coastal zones. A more detailed geographic representation at the city or region level would likely improve results for large or geographically diverse countries.

Physical geography. Haversine distance measures straight-line distance between centroids on a sphere. It ignores physical barriers that affect dialect diffusion: mountain ranges (Alps, Andes, Caucasus), bodies of water (Mediterranean, Atlantic), and political borders that restrict contact. Lebanon and Palestine are 232 km apart by Haversine distance but have had limited direct contact for decades. A road distance or contact intensity measure might better predict linguistic similarity.

Small sample of families. Seven families is a small sample for a correlation claim. Our central result is a non-correlation, so the risk runs the other way than usual: a real r to recovery relationship could exist and stay hidden at $n=7$. The signed- r

trend ($\rho=-0.71$, $p=0.07$) is suggestive but not significant, and German (5 varieties) and Portuguese (3 varieties) carry little weight on their own. A larger panel of families, and more varieties per family, would test whether even the weak sign signal holds.

Protocol mismatch. Arabic is evaluated on an external benchmark (NADI 2020), while all other languages use intrinsic CC-TLD LOO. These are not directly comparable. The NADI benchmark uses informal social media text from a fixed test set with human-verified labels. Intrinsic LOO uses the same noisy CC-TLD data for both training and testing. Recovery numbers across languages should therefore be interpreted with care: a 63% recovery for Arabic on a hard external benchmark is not the same experiment as a 57% recovery for German on its own held-out web text. This mismatch is also a reason not to over-read the cross-family correlation.

Model architecture. Character n -gram language models are simple and fast but miss word level and semantic variation. Many dialectal differences are lexical, a word choice that marks a speaker as Jordanian or Kuwaiti, and n -gram models capture these only through partial character overlap. Neural models trained on the same geographic data would likely achieve higher absolute performance and might show different $|r|$ behavior, particularly for families where lexical rather than phonological variation is the primary signal.

Coverage of world languages. We test seven language families, all using Latin, Arabic, or Cyrillic script. The framework has not been tested on tone languages, agglutinative families, or languages with complex morphology such as Turkish, Swahili, or Mandarin Chinese. The relationship between geographic proximity and linguistic similarity may be weaker or stronger in these families, and the register confound (colloquial vs. formal) may manifest differently.